

2(a)(i)	$t = 1.8 / 4.9$ $= 0.37 \text{ s}$	A1
2(a)(ii)	$v = u + at$ $= 9.81 \times 0.37$	C1
	$= 3.6 \text{ m s}^{-1}$	A1
2(a)(iii)	$v^2 = 3.6^2 + 4.9^2$	C1
	$v = 6.1 \text{ m s}^{-1}$	A1
	$\theta = \tan^{-1} (3.6 / 4.9)$ $= 36^\circ$	A1
2(b)(i)	$\rho = m / V$	C1
	$V = \frac{4}{3} \pi r^3$	C1
	$\rho = 0.017 / \left[\frac{4}{3} \pi \times (0.016 / 2)^3 \right]$ $= 7900 \text{ kg m}^{-3}$	A1
2(b)(ii)	$(E =) \frac{1}{2}mv^2$	C1
	$(E =) \frac{1}{2} \times 0.017 \times 4.9^2 = 0.20 \text{ (J)}$	A1

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Question	Answer	Marks
2(b)(iii)	$k = F/x$ or $k = \text{gradient}$	C1
	e.g. $k = 6.4 / 10 \times 10^{-2}$ $= 64 \text{ N m}^{-1}$ (allow 63–65 N m^{-1})	A1
2(b)(iv)	$E = \frac{1}{2}kx^2$ or $E = \frac{1}{2}Fx$ and $F = kx$	C1
	$x_0 = [(2 \times 0.20) / 64]^{0.5}$ $= 0.079 \text{ m}$ or 0.080 m	A1
2(c)(i)	same elastic potential energy / same (initial) kinetic energy and (polystyrene ball has) smaller mass (so greater speed) or same (average) force and (polystyrene ball has) smaller mass, (so greater average acceleration so greater speed)	B1
2(c)(ii)	(for the polystyrene ball there is) less (average vertical) acceleration / smaller (average vertical component of) <u>resultant</u> force (so takes longer time to reach ground)	B1